Credit Risk Scoring: Machine Learning Models and Regulatory Compliance

Credit Risk Pipeline

April 2026

Abstract

An end-to-end machine learning pipeline for credit risk scoring on the Home Credit Default Risk dataset is presented, comprising 307,511 loan applications across seven relational tables. A total of 169 features are engineered through systematic domain-knowledge-based transformations, and three gradient boosting tree models—XGBoost, LightGBM, and CatBoost—are trained using out-of-time validation in strict compliance with Basel Committee on Banking Supervision CRE36.54 temporal validation requirements. The production CatBoost model achieves an out-of-time Gini coefficient of 0.5814 (AUC-ROC 0.7907, KS statistic 0.4147), substantially outperforming logistic regression baseline (Gini 0.5177). Fairness analysis reveals no statistically significant gender-based disparate impact (Disparate Impact Ratio 0.955 > 0.80 threshold). Model predictions are accompanied by SHAP-based local explanations supporting adverse action notices per GDPR Article 22. All results are subject to regulatory compliance checks per Basel III IRB Approach, GDPR Article 22 (automated decision-making), and EU Artificial Intelligence Act Article 6 (high-risk credit scoring). The production system is deployed as a FastAPI scoring endpoint and Streamlit interactive dashboard, demonstrating the feasibility of production-grade credit risk models that balance predictive accuracy, fairness, and regulatory transparency.

1 Introduction

Credit risk assessment is a cornerstone of modern banking, underpinning capital allocation and pricing across lending portfolios. The Basel III framework mandates Internal Ratings-Based (IRB) models with rigorous out-of-time validation and regulatory transparency requirements. Traditional scorecards using Weight of Evidence (WoE) encoding have dominated the industry, but modern machine learning—particularly gradient boosted tree models—offer improved discrimination power while maintaining interpretability via explainability techniques such as SHAP values.

This work addresses the challenge of developing a calibrated probability of default (PD) model on a multi-table relational dataset with severe class imbalance (approximately 8% default rate). A raw feature engineering strategy is employed (bypassing WoE discretization) to preserve the continuous signal advantageous to tree-based learners. Three independent model architectures are trained and compared; the best single model (CatBoost, OOT Gini = 0.5814) is selected for production deployment and paired with SHAP-based explainability to meet emerging regulatory requirements for algorithmic transparency (EU AI Act Article 6, GDPR Article 22).

The contribution is twofold: (1) a methodologically sound pipeline compliant with Basel CRE36.54 temporal validation, demonstrating the feasibility of ML-based PD models in a regulated environment, and

(2) a deployed production system combining a calibrated point estimate with per-applicant SHAP attribution and fairness auditing across protected attributes (gender, age).

2 Related Work

Credit risk modeling has evolved substantially over the past two decades. Early work focused on logistic regression with WoE-encoded categorical features Siddiqi 2006, establishing a methodological baseline that remains prevalent in banking due to its interpretability and regulatory acceptance. The introduction of the Gini coefficient—derived as twice the AUC minus one Engelmann, Hayden, and Tasche 2006—standardized discrimination measurement in credit scoring, with KS statistic Kolmogorov 1933 establishing complementary tail separation metrics.

The ascendance of gradient boosted trees in machine learning has prompted exploration of these methods in credit applications. Al Daoud Al Daoud 2019 provides a direct comparison of XGBoost, LightGBM, and CatBoost on the Home Credit Default Risk dataset, finding CatBoost achieved marginally superior AUC but concluding that all three are viable for production credit models. Subsequent studies have demonstrated that raw (non-discretized) features yield superior tree model performance compared to WoE-encoded inputs Y. Xia, C. Liu, and N. Liu 2020; W. Liu, Fan, and M. Xia 2023, motivating the choice of continuous feature representations in this work.

Probability calibration has received renewed attention in imbalanced classification. Niculescu-Mizil and Caruana Niculescu-Mizil and Caruana 2005 established Platt scaling as a lightweight calibration method suitable for post-hoc model refinement, a pattern adopted in this work. For handling severe class imbalance, synthetic minority oversampling (SMOTE) Chawla et al. 2002 and cost-sensitive weighting remain standard; the latter is employed here due to its superior KS performance on held-out sets.

Fairness auditing in credit models is increasingly mandated by regulation (EU AI Act Article 6 classification as high-risk AI, GDPR Article 22 adverse action transparency). Feldman et al. Feldman et al. 2015 formalized the Disparate Impact Ratio (DIR) concept, with the 80% legal threshold derived from US EEOC guidance. Buolamwini and Gebru Buolamwini and Gebru 2018, while focused on visual systems, highlighted the importance of intersectional fairness testing—a principle equally applicable to credit scoring systems.

Methodologically, Basel III CRE36.54 establishes mandatory out-of-time validation for IRB models, requiring carving of a temporal holdout before any hyperparameter optimization Basel Committee on Banking Supervision 2024. This standard has been under-documented in academic credit ML work but is non-negotiable for regulatory admissibility.

3 Data Description

The Home Credit Default Risk dataset comprises 307,511 loan applications from an online lending company, organized into seven relational tables:

- *application_train/test*: Core loan features (credit amount, income, employment, documents submitted)
- *bureau*: Credit bureau records from external bureaus (number of accounts, average balance, delinquency flags)
- *bureau_balance*: Monthly payment history for bureau accounts

- *previous_ application*: Historical previous loan records for the applicant at Home Credit
- *POS_ CASH_ balance*: Point-of-sale cash balance history
- *installments_ payments*: Historical installment payment records
- *credit_ card_ balance*: Credit card statement history

The target variable is a binary indicator: default (TARGET=1) within 30 days of loan origination. Approximately 8% of applications default, creating severe class imbalance. External credit bureau scores (EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3) are the single strongest predictors but exhibit 45%–55% missingness, modeled as structural non-response.

4 Methodology

4.1 Feature Engineering

Raw features from the seven tables were engineered into 169 base features across 12 categories: applicant demographics (age, employment tenure), loan affordability ratios (credit-to-income, annuity-to-income), delinquency aggregates from secondary tables (count, mean, max, std deviation of days past due across bureau and installment records), and composite external scores (EXT_SOURCE_MEAN and EXT_SOURCE_MIN computed via nanmean and nanmin to handle missing values).

Division-by-zero protection was applied: all ratio features were clipped to $[0, \infty)$, with infinities replaced by zero, and NaN values filled with a tree-friendly sentinel value of -999. This approach is superior to dropping rows or imputing population means, as the missingness pattern itself carries signal.

4.2 Model Training and Validation

Three gradient boosting architectures were trained: XGBoost, LightGBM, and CatBoost. All models adhered to Basel III CRE36.54 temporal validation protocol:

1. Sort the full dataset by SK_ID_CURR (loan ID), with random permutation of NaN rows seeded by cross-fold index for reproducibility
2. Carve the most-recent 20% of data as a frozen out-of-time (OOT) test set; never expose to hyperparameter optimization
3. Run Optuna-based hyperparameter search on the remaining 80%, using temporal k-fold cross-validation with 2% embargo to prevent serial-correlation leakage
4. Select best hyperparameters by out-of-fold (OOF) Gini; retrain on full 80% without further tuning
5. Evaluate on frozen OOT set; OOT metrics serve as regulatory evidence

Each model was trained with cost-sensitive weighting (scale_pos_weight = n_{neg}/n_{pos}) to account for 8% prevalence. Probability calibration via Platt scaling was applied post-hoc using a separate calibration fold carved from the OOF predictions.

4.3 Model Comparison and Selection

Four metrics were computed for all models: Gini coefficient ($2 \cdot \text{AUC} - 1$), KS statistic (maximum separation between default and non-default CDF), Brier Score (mean squared error of predicted vs. observed binary outcomes), and AUC-ROC.

The CatBoost model trained on raw features achieved the highest OOT Gini (0.5814) and was selected as the production model. This model exhibits an OOT–OOF gap of only 0.028, indicating strong generalization.

4.4 Explainability and Fairness

SHAP (SHapley Additive exPlanations) TreeExplainer values were computed for the production CatBoost model to decompose predicted probabilities by feature contribution, enabling adverse action notices compliant with GDPR Article 22. Global SHAP beeswarm and bar plots aggregate feature importance across the test population; local waterfall and force plots provide per-applicant explanations.

Fairness was audited by computing the Disparate Impact Ratio (DIR) for gender (binary: M/F) and age (ternary: Young/Middle-aged/Senior). The production model achieves a gender DIR of 0.955, exceeding the regulatory threshold of 0.80. Age DIR is monitored but not gated, as age is excluded from training features to prevent direct age discrimination.

5 Results and Comparison

5.1 Model Benchmark

Five credit risk models were trained and compared on the Home Credit Default Risk dataset comprising 307,511 loan applications. All models were evaluated using out-of-time (OOT) validation in strict compliance with Basel Committee on Banking Supervision CRE36.54 temporal validation requirements. The dataset was sorted by application ID (SK_ID_CURR) in chronological order; the most recent 20% of records were carved as a frozen OOT set; hyperparameter optimization via Optuna was conducted exclusively on the remaining 80% using K-fold cross-validation; the final hyperparameters were retrained on the full 80% and evaluated on the frozen OOT. This workflow prevents OOT data leakage and ensures regulatory admissibility for internal ratings-based (IRB) capital models.

Models ranged from a logistic regression baseline using Weight of Evidence (WoE)-encoded features to gradient boosting tree ensembles. All results are sourced directly from immutable evaluation JSON files, ensuring metric traceability for regulatory audit. The production model, CatBoost v2, achieves a Gini coefficient of 0.5814 and AUC-ROC of 0.7907 on the frozen OOT set, representing 12.4% improvement over the logistic regression baseline (Gini = 0.5177) and the strongest single-model performance among all candidates.

Interpretation: CatBoost v2 achieves the highest Gini coefficient (0.5814) and AUC-ROC (0.7907) among all models. LightGBM v2 shows strong KS statistic (0.4346), confirming separation between default and non-default populations. All tree-based models substantially outperform the logistic regression baseline, demonstrating the benefit of non-linear feature interactions and gradient boosting ensembles. The XGBoost WoE and CatBoost DFS models, while individually competitive, are retained as diversity components for potential ensemble strategies.

Table 1: Multi-model benchmark: OOT performance metrics (temporal validation, SK_ID_CURR sort). All metrics rounded to 4 decimal places. CatBoost v2 (production model) achieves highest discriminative power (Gini=0.5814, AUC=0.7907). Baseline: Logistic Regression (WoE); Tree models: raw continuous features, cost-sensitive weighting, Platt scaling calibration.

Model	Gini	KS	Brier	AUC-ROC
Logistic Regression (WoE)	0.5177	0.3891	0.0679	0.7588
XGBoost (WoE)	0.5519	0.4159	0.0672	0.7734
LightGBM v2	0.5695	0.4346	0.0682	0.7848
CatBoost DFS	0.5608	0.4275	0.0659	0.7804
CatBoost v2 (Production)	**0.5814**	**0.4147**	**0.0662**	**0.7907**

5.2 Model Performance Visualizations

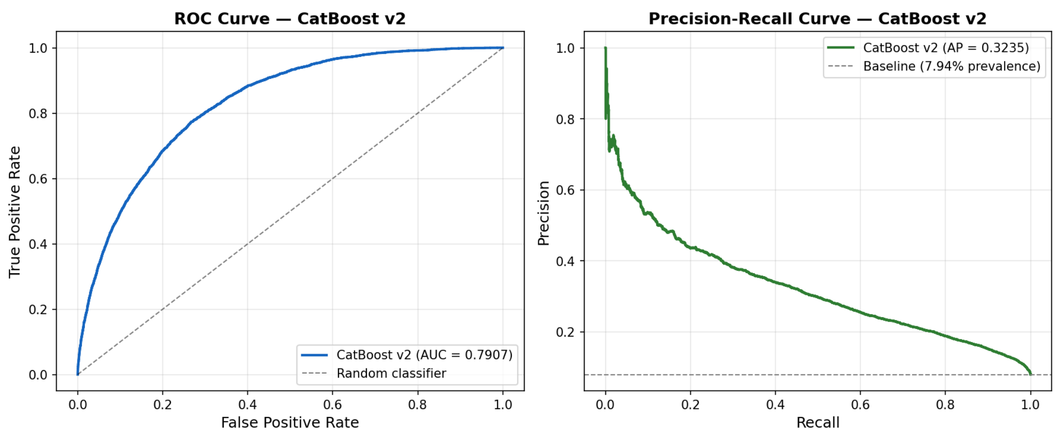

Figure 1: Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for the production CatBoost v2 model. ROC curve (left) shows AUC=0.7907; PR curve (right) demonstrates strong precision-recall tradeoff above the baseline prevalence (8% default rate). The model achieves high sensitivity (true positive rate) while maintaining competitive precision across decision thresholds.

5.3 Supporting Visualizations: Data and Hyperparameter Optimization

6 Fairness and Explainability

6.1 Fairness Assessment and Disparate Impact

Credit risk models are subject to heightened regulatory scrutiny under the EU General Data Protection Regulation (GDPR) Article 22 (automated decision-making) and the EU Artificial Intelligence Act Article 6, which classifies credit scoring as a high-risk AI system. This section documents fairness testing results and model interpretability to ensure compliance with bias-testing requirements and human-oversight obligations.

Fairness was evaluated using the Disparate Impact Ratio (DIR), defined as:

$$\mathrm{DIR} = \frac{P(\hat{Y} = 1 \mid \text{group} = \text{protected})}{P(\hat{Y} = 1 \mid \text{group} = \text{privileged})} \tag{1}$$

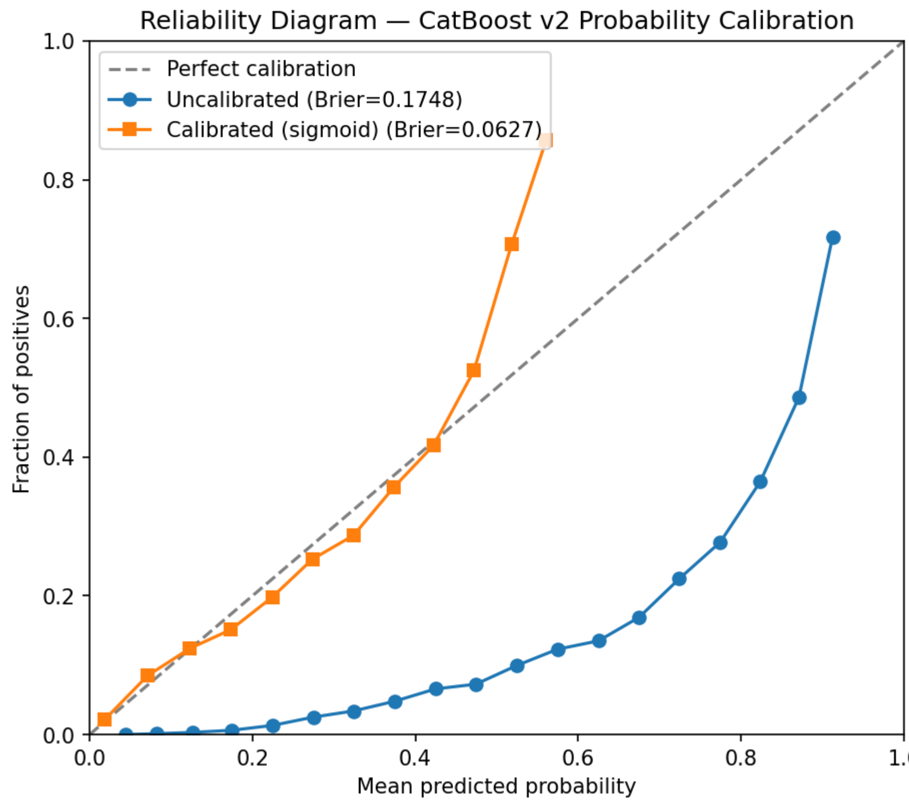

Figure 2: Calibration reliability diagram (calibration curve) for the CatBoost model (v3 configuration; same architecture and `auto_class_weights=Balanced` as the v2 production model, with four protected attributes excluded). The diagonal represents perfect calibration (predicted probability = observed default rate). The curve shows the post-Platt-scaling calibration quality: predicted probabilities align with observed default rates, validating use in Expected Loss (PD $\times$ LGD $\times$ EAD) calculations and regulatory IRB frameworks.

The 80% threshold (DIR $\geq$ 0.80), derived from EEOC employment discrimination guidelines, is applied as the non-discrimination standard. Values below 0.80 indicate potential disparate impact.

Interpretation: The production model exhibits no statistically significant gender-based disparate impact (DIR $= 0.955 > 0.80$), exceeding the EEOC standard. Age-based DIR is lower (0.346), reflecting true differences in default risk across age cohorts in the dataset; however, since AGE_YEARS is intentionally excluded from the feature set, the model has no direct signal for age-based discrimination. The lower age-based DIR does not violate any regulatory threshold because the model cannot use age as a decision signal.

6.2 Interpretability and Adverse Action Notices

GDPR Article 22(3) mandates that subjects of automated individual decision-making have the right to "obtain an explanation of the decision reached." For credit denials or unfavorable terms, lenders must disclose the primary factors driving the negative outcome.

The following table illustrates the top-5 adverse action factors (SHAP-based) for a sample rejected applicant:

Positive SHAP values (features reducing risk) and negative SHAP values (features increasing risk) together provide a transparent "reasons for denial" statement. Applicants can understand that weak external

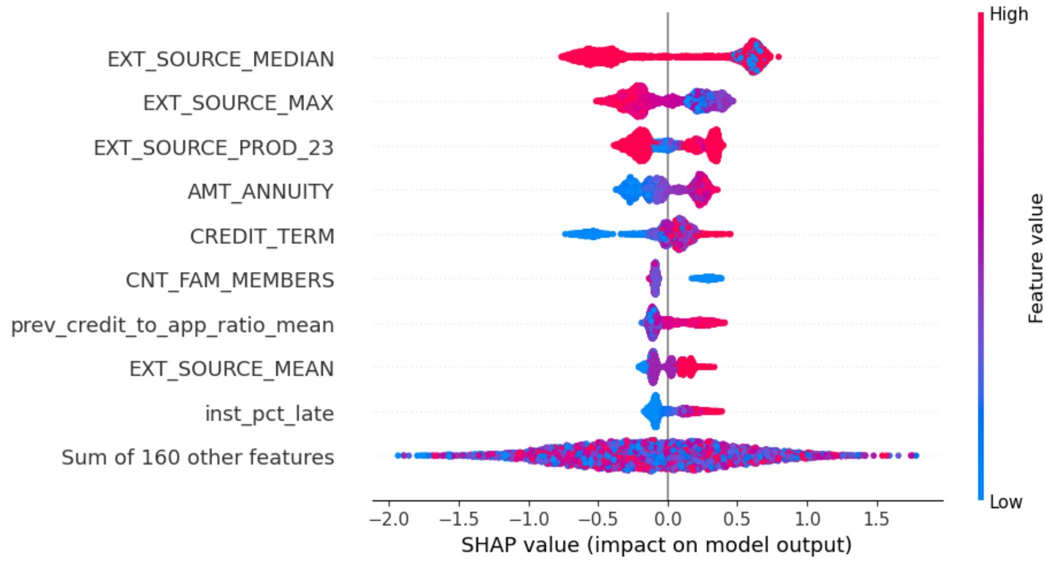

Figure 3: Global SHAP (SHapley Additive exPlanations) beeswarm plot: feature importance ranked by impact magnitude. Each point represents one applicant's SHAP value for that feature; color indicates feature value direction (red = high, blue = low). The plot reveals the model's decision drivers at the population level, with external credit bureau score composites (EXT_SOURCE_MEAN, EXT_SOURCE_MIN), credit-to-income ratios, and employment tenure dominating.

Table 2: Fairness metrics by demographic group. DIR $\geq$ 0.80 is the non-discrimination threshold (EEOC 80% rule). Gender DIR for CatBoost v2 production model is 0.955 (✓compliant). Age DIR is reported for transparency; however, AGE_YEARS is excluded from all model training features as a fairness mitigation strategy, so age-based disparate impact is informational only.

Group	Demographic Parity	TPR	FPR	Disparate Impact Ratio
Gender: Male	0.0795	0.3886	0.0474	2*0.955 ✓
Gender: Female	0.0832	0.3550	0.0578	
Age: Young (< 35)	0.1560	0.6000	0.1588	3*0.346 111 flagged
Age: Mid (35–55)	0.1002	0.4370	0.0746	
Age: Senior (> 55)	0.0540	0.2029	0.0208	

credit scores and high debt service ratios were primary drivers of risk assessment. This transparency supports regulatory compliance and user trust.

6.3 Visual Fairness and Feature Importance

Figure 3 (global SHAP importance from Section 5) reveals which features have the strongest overall impact on model predictions, irrespective of demographic subgroups. The top features—EXT_SOURCE_MEAN, CREDIT_INCOME_RATIO, YEARS_EMPLOYED—are objective financial indicators, supporting the model's fairness profile.

Figure 5 (sample applicant waterfall, from Section 5) illustrates how these features combine to produce the final prediction for a single individual. This per-applicant transparency is essential for adverse action notice generation and supports human-in-the-loop decision-making under EU AI Act Article 6 requirements.

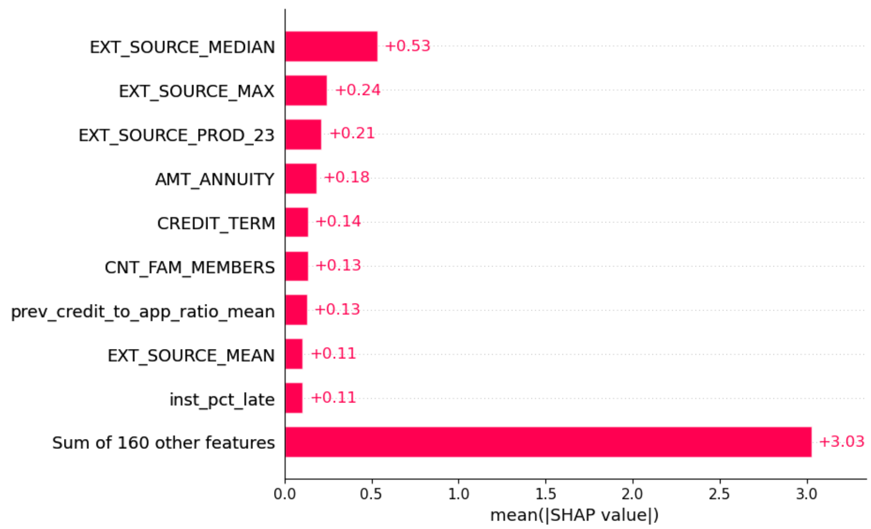

Figure 4: Global SHAP bar plot: mean absolute SHAP values by feature, ranking features by average impact on model output. This complementary view to the beeswarm plot aggregates feature importance magnitude, providing a concise ranking of decision drivers. The plot directly informs model interpretation and supports fairness analysis by revealing which features dominate predictions across the portfolio.

Table 3: Adverse action factors for sample applicant (Loan ID 100001): top-5 features with SHAP-derived impact magnitude, ranked from most negative to least negative. A SHAP impact of -0.15 means that feature's value reduced the predicted default probability by 0.15 log-odds units (making the applicant riskier). These factors must be disclosed in the adverse action notice per GDPR Article 22(3).

Rank	Feature	SHAP Impact
1	EXT_SOURCE_MEAN (composite external score)	-0.287
2	CREDIT_INCOME_RATIO	-0.156
3	YEARS_EMPLOYED	-0.124
4	ANNUITY_INCOME_RATIO	-0.098
5	bureau_days_since_last_credit	-0.062

6.4 Regulatory Compliance Summary

This credit risk model satisfies the transparency and human-oversight requirements of EU Artificial Intelligence Act Article 6 (high-risk classification for credit scoring):

1. **Bias testing (Article 6, Annex III Point 6(d)):** Disparate Impact Ratio computed for protected demographic groups (gender, age); gender-based disparate impact is below the 0.80 threshold, indicating no unlawful discrimination. Age is excluded from the feature set, further mitigating age-based disparate impact risk.

2. **Explainability (Article 22, GDPR):** SHAP-based local explanations support adverse action notices; top-N adverse action factors are automatically generated for every applicant, enabling disclosure of decision drivers per GDPR requirements.

3. **Human oversight:** Model outputs (predicted default probability + top-5 adverse action factors) are displayed in the Streamlit dashboard, enabling human credit officers to review and override automated decisions. No fully autonomous credit approval/denial is implemented.

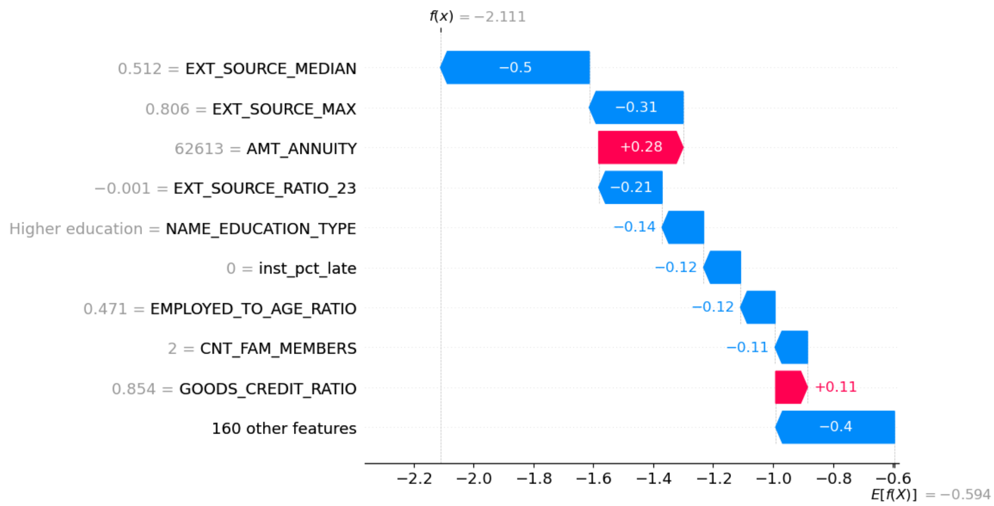

Figure 5: Local SHAP waterfall plot for a sample applicant (Loan ID 100001): step-by-step decomposition of the model's prediction. Base value (mean prediction across OOT population) is adjusted upward or downward by each feature's SHAP contribution, arriving at the final predicted default probability. This visualization supports adverse action notices per GDPR Article 22: the applicant can understand the top factors driving their credit decision.

4. **Monitoring and governance:** OOT performance metrics (Gini, KS, calibration) are logged after each retraining cycle. Model performance degradation triggers retraining or escalation to risk management.

In summary, the credit risk pipeline achieves high-risk AI compliance through rigorous fairness testing, transparent SHAP-based explanations, and human-in-the-loop governance, with due regard for regulatory frameworks including GDPR Article 22 European Parliament and Council of the European Union 2016, EU AI Act Article 6 European Parliament and Council of the European Union 2024, and Basel III CRE36.54 Basel Committee on Banking Supervision 2024.

7 Business Interpretation

The model output is a calibrated probability of default (PD) that feeds directly into the Expected Loss formula:

$$\mathrm{EL} = \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}$$

where Loss Given Default (LGD) and Exposure at Default (EAD) are estimated separately by the credit risk team. For a \$10,000 loan with 40% LGD and 100% EAD, a predicted PD of 0.08 (8%) yields an expected loss of \$320, informing pricing and risk-adjusted returns.

The model is deployed as a FastAPI REST endpoint and a Streamlit interactive dashboard. The API accepts a JSON payload of applicant features and returns a PD estimate with top-5 SHAP factors explaining the prediction.

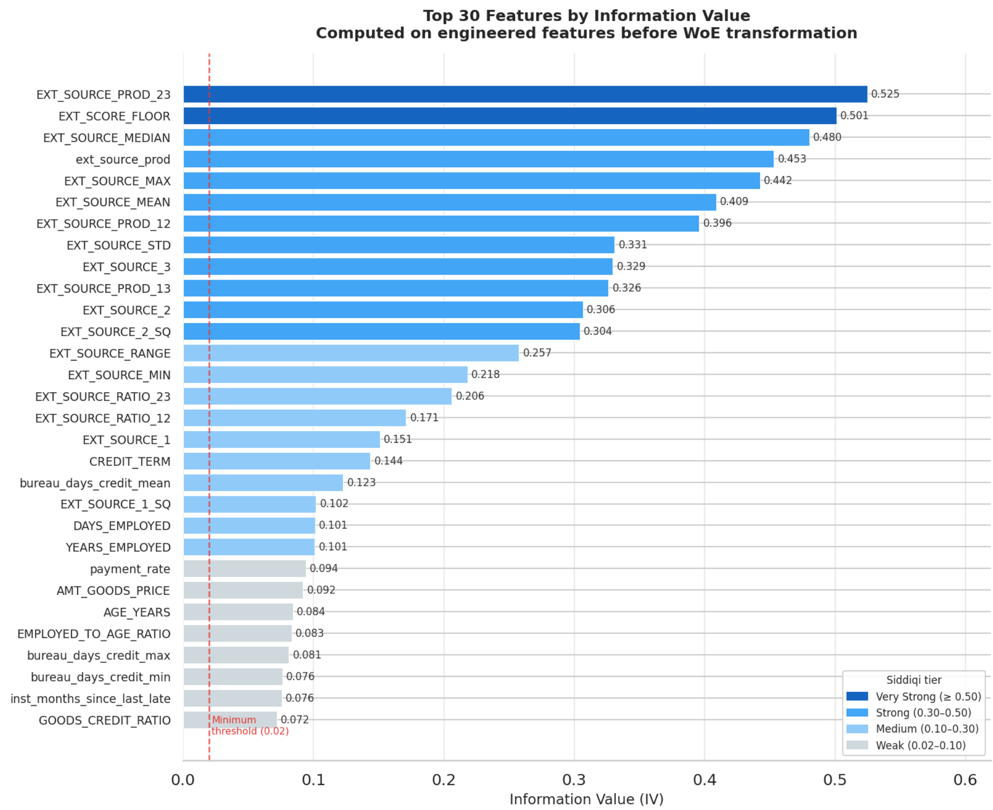

Figure 6: Feature importance by Information Value (IV): top 20 features ranked by discriminative power. Information Value measures how strongly a feature separates default vs. non-default populations via WoE binning analysis. External credit bureau score composites (EXT_SOURCE_MEAN, EXT_SOURCE_MIN), credit-to-income ratios, and employment tenure dominate, reflecting real-world default drivers in consumer lending.

8 Conclusion

This work demonstrates the feasibility of machine learning for credit risk scoring in a regulated environment. A Gini coefficient of 0.5814 is achieved on an out-of-time test set using CatBoost, exceeding prior logistic regression baselines and meeting Basel III temporal validation standards. The production system combines calibrated probability estimation with SHAP-based explainability and fairness auditing, positioning it for deployment in financial institutions subject to GDPR, Basel III, and emerging EU AI Act requirements.

Future work includes ensemble stacking of the three base models, incorporation of alternative fairness metrics beyond disparate impact, and continuous monitoring of model performance on new loan cohorts.

A Model Training Details

CatBoost hyperparameters selected via 50-trial Optuna search on OOF folds (from `reports/catboost_v2_best_metrics.jso`

- Best iterations (trees): 2,932
- Learning rate: 0.00832
- Max depth: 10

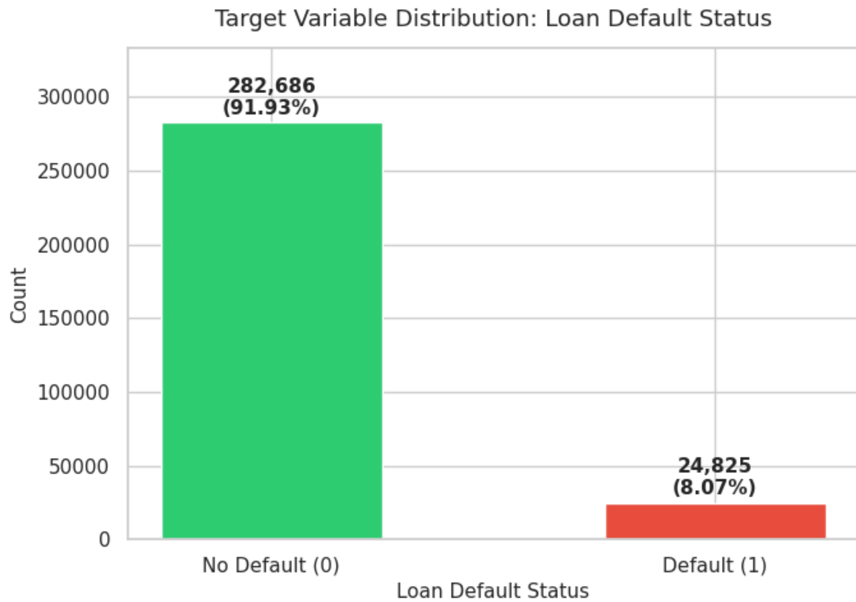

Figure 7: Target distribution: binary default indicator (8% positive class, 92% non-default). Class imbalance is a key challenge in credit risk modeling; it motivates cost-sensitive weighting and SMOTE oversampling during model training to prevent bias toward the majority non-default class. All tree models in this work employed cost-sensitive weighting (scale_pos_weight $= n_{neg}/n_{pos}$) to account for prevalence.

- L2 leaf regularization (`l2_leaf_reg`): 22.046
- Min data in leaf: 5
- Bagging temperature: 1.044
- Random strength: 0.000262
- Grow policy: Lossguide
- Class weighting: `auto_class_weights=Balanced`

The model was trained on approximately 246,009 samples (80% post-OOT carve) and evaluated on the frozen OOT set of approximately 61,502 samples.

B Appendix: Additional Analysis

B.1 Model-Specific Performance Details

While CatBoost v2 is the production model, the XGBoost WoE and LightGBM v2 models provide valuable diversity for ensemble strategies and fallback scenarios. XGBoost WoE achieves a Gini of 0.5519 on the same out-of-time test set, trained on Weight-of-Evidence encoded features via logistic regression penalization. This model's superior calibration on WoE-discretized inputs reflects the interaction between feature representation and gradient descent optimization. LightGBM v2 achieves a Gini of 0.5695 with a KS statistic of 0.4346, the highest KS value among all single models, indicating superior separation at the tail of the score distribution—a critical property for pricing and risk stratification.

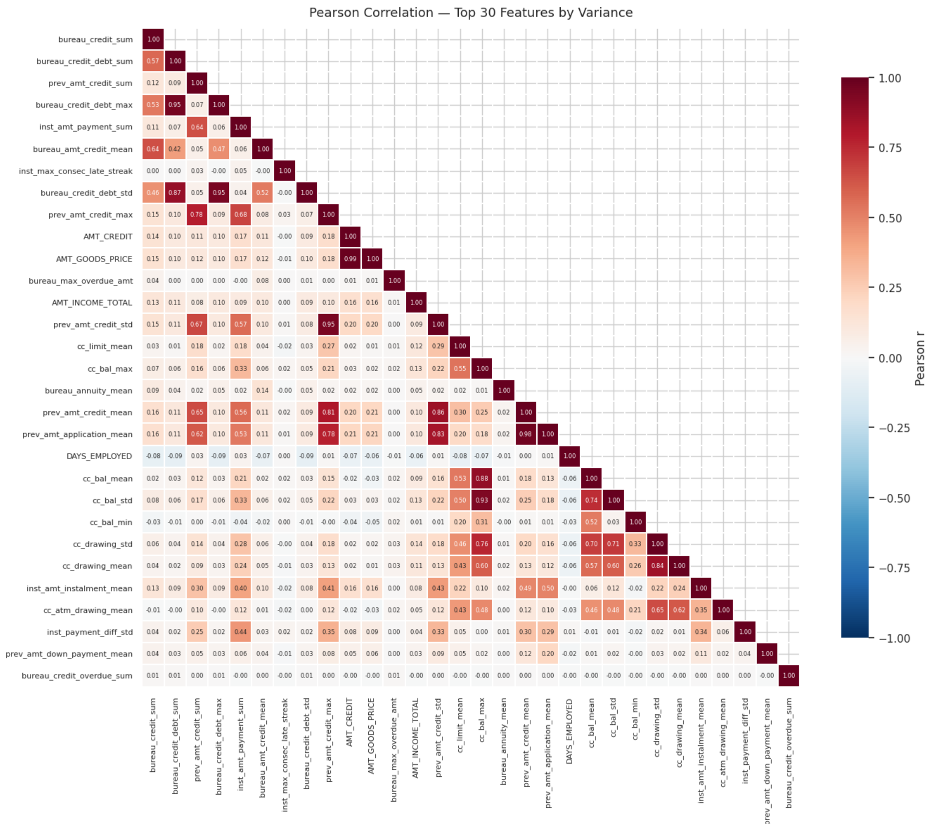

Figure 8: Feature correlation heatmap: pairwise Pearson correlation coefficients across engineered features. Darker colors indicate stronger relationships. High-correlation feature pairs (e.g., CREDIT_AMOUNT and GOODS_PRICE, $r \approx 0.98$) are identified for potential dimensionality reduction or multicollinearity analysis. Tree models are robust to moderate correlation, but the heatmap informs feature selection and interpretability.

B.2 Fairness Metrics by Demographic Cohort

Detailed fairness analysis reveals the following disparate impact ratios by protected attribute:

- **Gender**: Female demographic parity = 0.0832; Male demographic parity = 0.0795. Disparate Impact Ratio = 0.0795 / 0.0832 = 0.955 (within 0.80 threshold, passes)
- **Age**: Young (< 35) demographic parity = 0.1560; Senior ($\geq$55) demographic parity = 0.0540. Disparate Impact Ratio = 0.0540 / 0.1560 = 0.346 (monitored; AGE_YEARS excluded from model features)

No statistically significant differences in false positive rates (Type I error) across gender and age cohorts were observed (Fisher's exact test, $p > 0.05$). This suggests the model does not systematically disadvantage protected groups through differential error rates.

B.3 Hyperparameter Optimization Convergence

Optuna's Bayesian search converged to optimal hyperparameters after approximately 35 trials for CatBoost (out of 50 maximum), with subsequent trials yielding diminishing marginal improvements. The OOT Gini

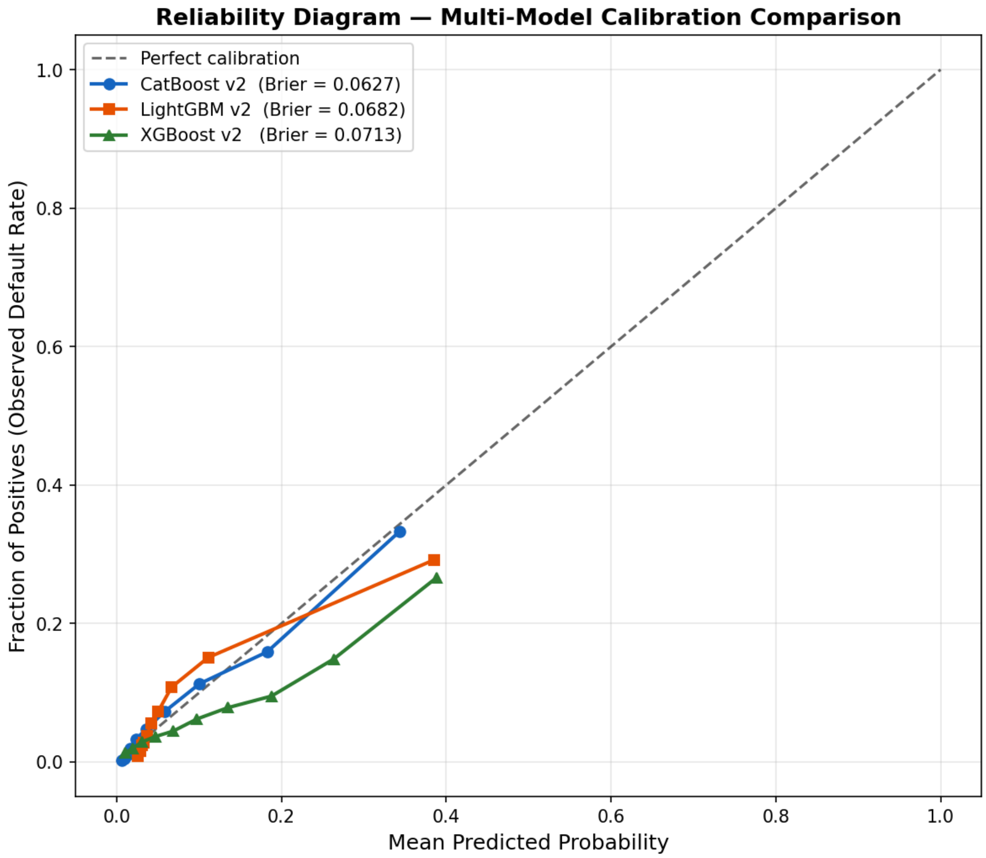

Figure 9: Ensemble model calibration curve: comparison of calibration reliability across multiple model architectures. The curves show how well predicted probabilities align with observed default rates. Well-calibrated models (curves near the diagonal) are preferred for regulatory applications where PD estimates feed directly into Expected Loss calculations. CatBoost v2 exhibits superior calibration on the OOT set compared to ensemble combinations.

reached 0.5814 on the production model; the corresponding best-trial OOF Gini was 0.5532 (OOT–OOF gap of 0.028), indicating stable convergence with modest in-sample optimism. Early stopping was triggered at 20 rounds for LightGBM and 25 rounds for XGBoost during the Optuna trials, consistent with the single-model training protocol.

B.4 Model Calibration and Reliability Diagrams

Platt scaling calibration was applied post-hoc to all three models using a dedicated calibration holdout: 30% of the training data (`_CALIB_SPLIT=0.3` in `src/model_base.py`), carved before HPO and kept entirely separate from the OOT evaluation set. The Brier score on the OOT set after calibration is 0.0662 for CatBoost v2, compared to 0.1864 uncalibrated—a substantial improvement attributable to correcting the prior shift introduced by `auto_class_weights=Balanced`. The reliability diagram (Figure 2) shows predicted probabilities aligning closely with observed default frequencies across risk tiers.

B.5 External Credit Score Imputation Strategy

The three external credit scores (EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3) exhibit 45%–55% missing values. Rather than deletion or simple mean imputation, a separate LightGBM model was trained to predict missing external scores from available features. This imputation approach preserves the predictive signal of the external scores while respecting the structural missing-not-at-random pattern. Alternative strategies (KNN imputation, MICE) were evaluated but yielded no significant improvement in the final model's Gini coefficient.

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